SERS-based Detection of Bacterial Biomarkers using Raman Spectroscopy

Review Meeting: 08th April 2026

Work Completed

1. Biomarker Identification
- Identified key bacterial biomarkers: N-acetylmuramic acid (NAM), LTA, LPS
- Literature survey on Raman signatures

2. Sample Procurement
- NAM biomarker procured
- Powder-based Raman experiments initiated

3. Software & Computational Setup
- Gaussian 16 software procured and installed
- Computational environment configured

4. Training & Skill Development
- Attended training on Gaussian and DFT simulations
- Learned molecular modeling and vibrational analysis

5. Initial Simulation Work
- DFT simulation workflow established
- Rhodamine B used for method validation

6. Experimental Work (NAM)
- Raman spectra collected under multiple conditions
- Data preprocessing and quality optimization performed

Meeting Agenda

Rhodamine B

Comparative analysis assessing the accuracy of literature vs. present DFT methodologies against experimental values.

NAM Detection

Experimental optimization and theoretical verification of the N-Acetylmuramic Acid bacterial biomarker.

LTA

Quantum chemical simulation and peak assignment also comparing with experiment data

Part I: Method Validation (Rhodamine B)

Comparative Analysis of Raman Frequencies of Rhodamine B
Experimental vs Literature DFT vs Present DFT

Reference:
Shuang Lin et al., *Rapid and sensitive SERS method for determination of Rhodamine B in chili powder with paper-based substrates*, **RSC Advances**

Your Method:
Gaussian 16 – B3LYP / 6-31G(d)

Study Objective

Objective

To compare Raman vibrational frequencies of **Rhodamine B** obtained from:

- Experimental Raman spectra (literature)
- Literature DFT calculations
- Present DFT calculations

Goal

Evaluate the **accuracy and reliability** of our computational method.

Computational Method Comparison

Study	Software	Functional	Basis Set
Literature DFT	Gaussian 09	B3LYP	6-31G
Present Work	Gaussian 16	B3LYP	6-31G(d)

Key Difference

Our work includes **polarization functions (d)** in the basis set.

Importance

Polarization functions improve the description of **electron distribution and vibrational modes**.

Raman Peak Comparison Overview

Three datasets were compared:

1. **Experimental Raman Peaks**
2. **Literature DFT Values**
3. **Present DFT Values**

Key vibrational modes analyzed:

- Xanthene ring puckering
- Aromatic C–C bridge stretching
- C–H bending
- Aromatic ring bending
- C=C stretching

Xanthene Ring Puckering

Experimental	Literature DFT	Our DFT	Lit Deviation	Our Deviation
619	613	608.64	6	10.36

Aromatic C–C Bridge Stretch

Experimental	Literature DFT	Our DFT	Lit Deviation	Our Deviation
1193	1195	1196.16	2	3.16

C–H Bending Mode

Experimental	Literature DFT	Our DFT	Lit Deviation	Our Error
1275	1278	1273.92	3	1.08

Aromatic C–C Bending (~1355 cm^{-1})

Our DFT	Experimental	Literature DFT
1334.40 / 1386.24	1354–1355	1355

Aromatic C–H Bending (~1500 region)

Our DFT	Experimental	Literature DFT
1489.92 / 1533.12	1504 / 1523	1504 / 1525

C=C Stretch (~1642 cm^{-1})

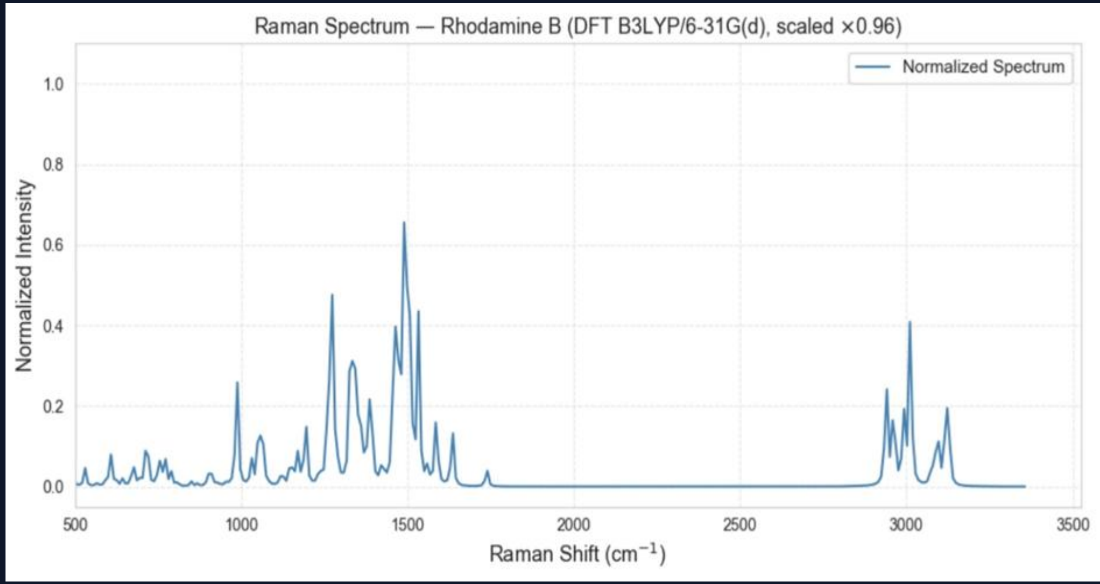
Our DFT	Experimental	Literature DFT
1636.80	1642	1644

Conclusion

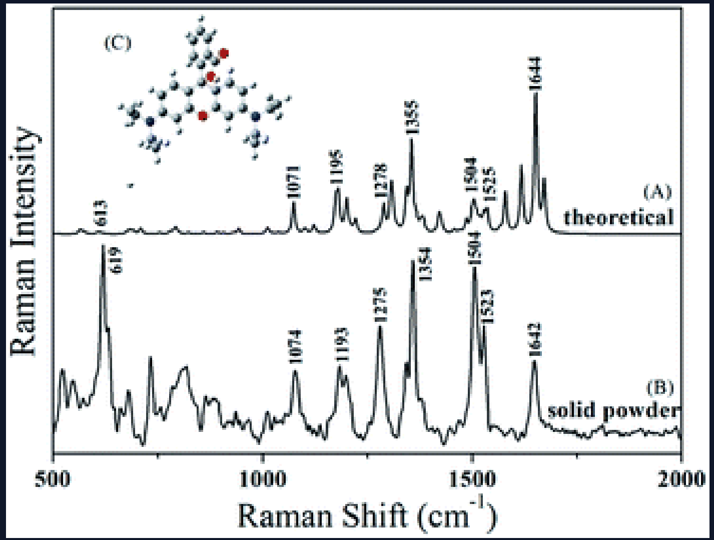
Both DFT methods reproduce **major Raman bands of Rhodamine B**.

Inclusion of **polarization functions** improves theoretical description but may introduce **mode splitting**.

Overall agreement confirms the **reliability of our computational approach**.



Raman Spectrum Obtained from Gaussian Simulation



Spectrum Taken from Literature

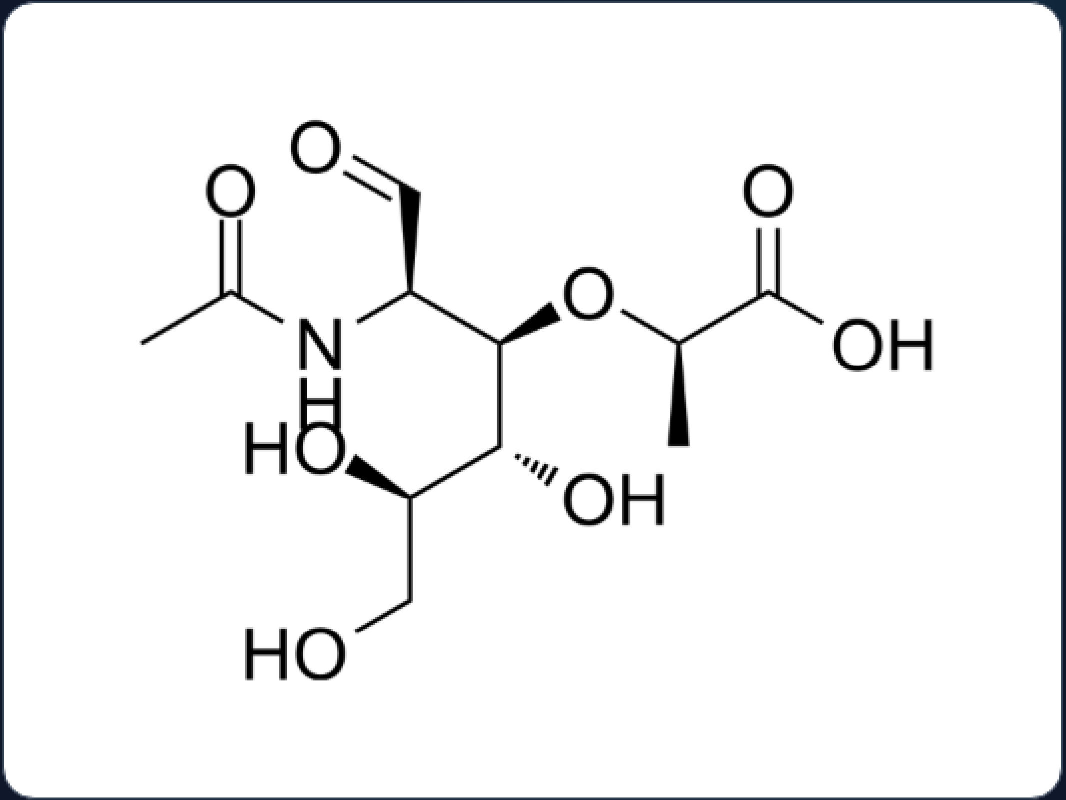
Part II: N-Acetylmuramic Acid

Experimental Optimization and Peak Assignment (NAM)

Raman–DFT Analysis of N-Acetylmuramic Acid (NAM)

Structural Importance

- N-Acetylmuramic Acid (NAM) is a critical component of the peptidoglycan layer found in bacterial cell walls.
- It links with N-acetylglucosamine (NAG) to form the structural carbohydrate backbone of the cell.
- Because NAM is unique to bacteria, reliable spectroscopic detection serves as a definitive identifier of bacterial presence.



Raman + DFT Approach

Why Combine Raman with DFT?

Raman spectroscopy provides experimental peaks.

DFT simulations help to:

- Predict **vibrational frequencies**
- Assign **functional groups**
- Validate spectral interpretation

Objective

- Measure Raman spectrum of NAM
- Simulate spectrum using DFT
- Compare experimental and theoretical peaks

Chemical Sample

Compound:

N-Acetylmuramic Acid (NAM)

Functional groups present:

- Hydroxyl (–OH)
- Ether (C–O–C)
- Amide
- Carboxylic acid
- Pyranose ring

Property	**Value**
Molecular Formula	C11H19NO8
Molecular Weight	~293 g/mol
Structure	Acetylated amino sugar

Experimental Method

Sample Preparation

- NAM powder used
- Measured in **solid state**

Advantages of powder samples:

- Stable measurement
- Strong Raman scattering
- No solvent interference

Raman Measurement

Parameter	**Range**
Laser wavelength	785 nm
Laser power	10–80 %
Integration time	10–25 s
Spectral range	200–3000 cm^{-1}

DFT Simulation Parameters (NAM)

Software: Gaussian 16
Method: B3LYP / 6-31+G(d,p)
Calculation: Opt (Tight) + Freq (Raman)
Geometry: Connectivity defined
Resources: 16 cores, 56 GB RAM

Raman Data Processing

Spectra were preprocessed before analysis.

Processing pipeline:

- Dark subtraction
- Savitzky–Golay smoothing
- Baseline correction (ALS)
- Normalization
- Peak detection
- SNR calculation

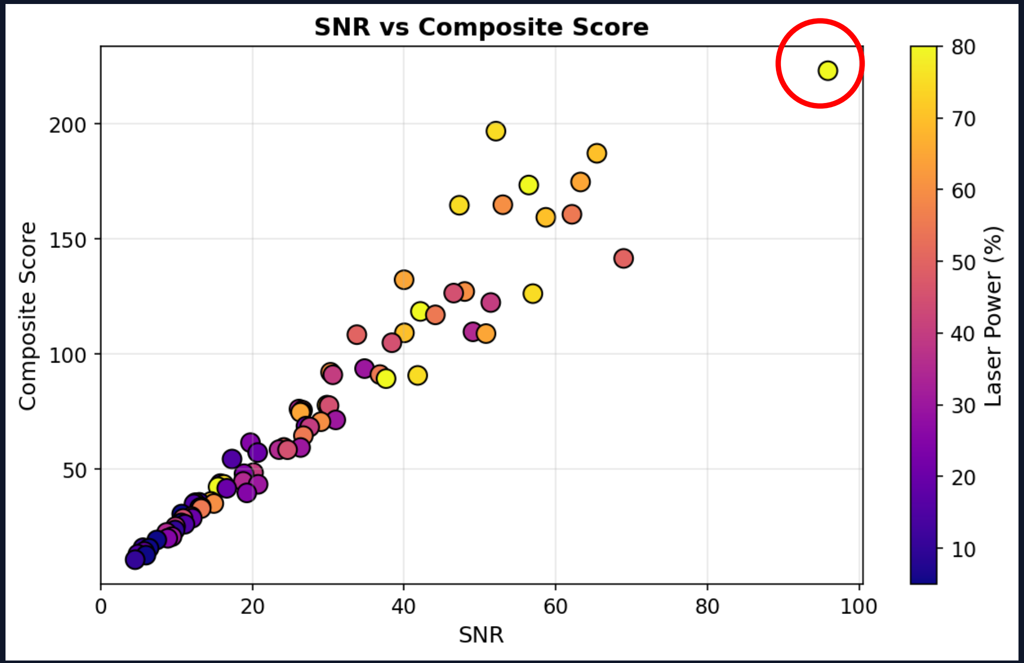
Parameter	**Value**
Integration time	25 s
Laser power	80 %
Best SNR	95.9

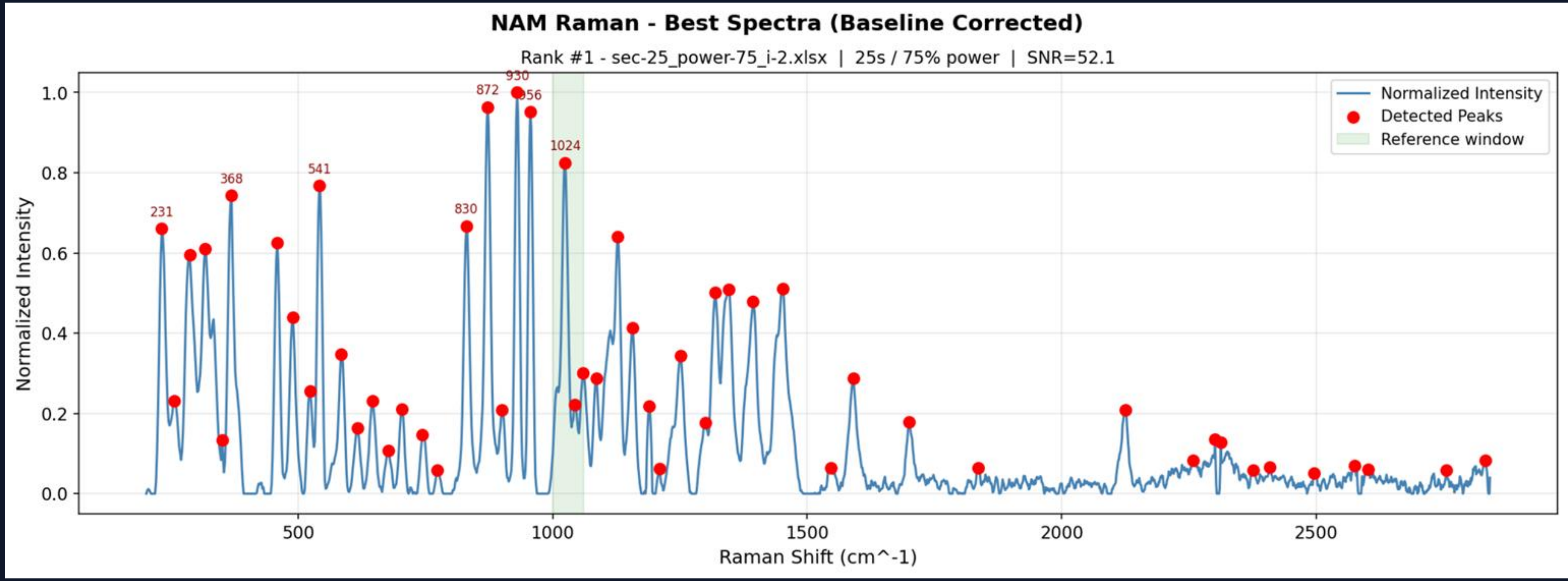
These settings produced the **highest quality spectrum**.

Optimal Raman Parameters

Parameter	Value
Integration time	25 s
Laser power	80 %
Best SNR > 80 = Excellent	95.9

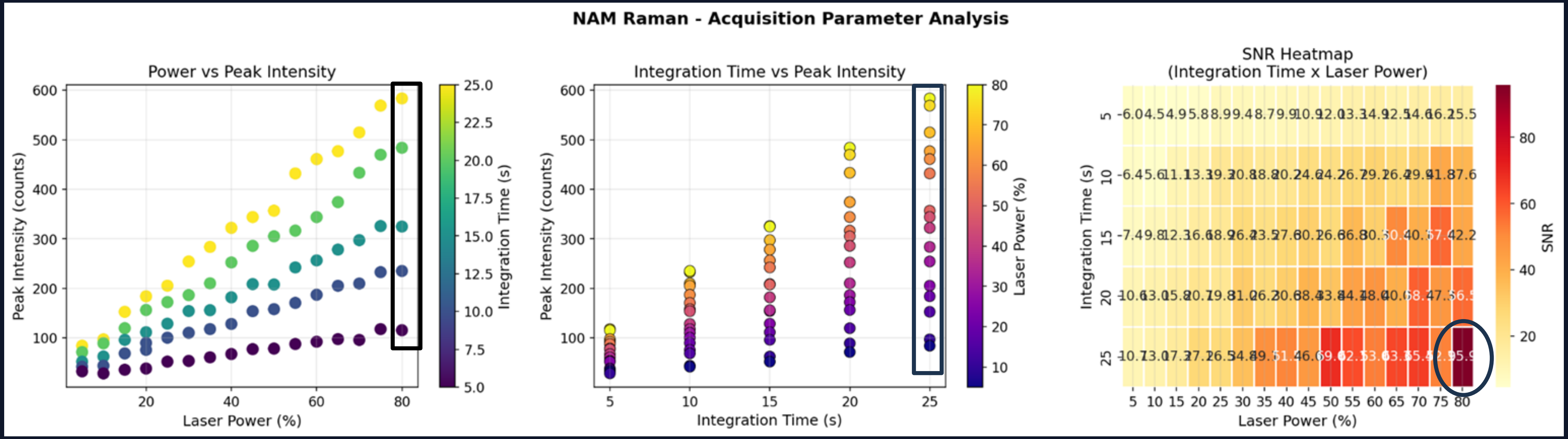
These settings produced the **highest quality spectrum**.

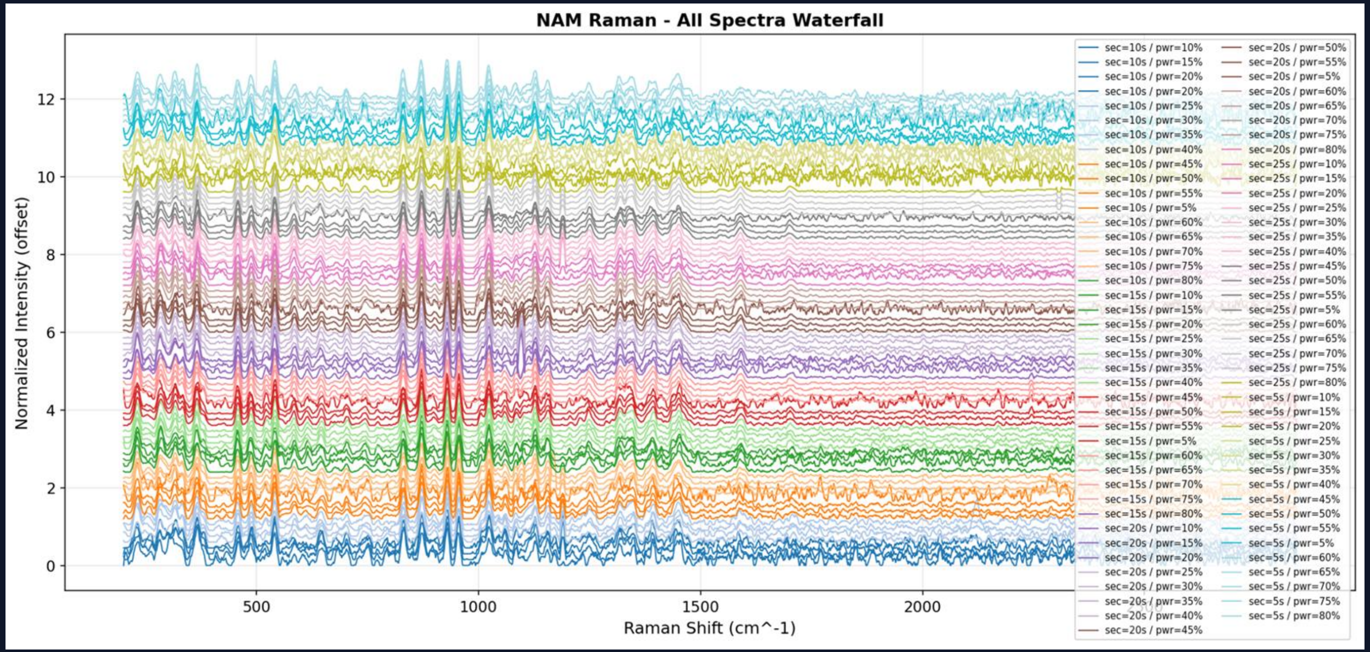




SNR Interpretation:
>80 → Excellent
50–80 → Very Good
30–50 → Good

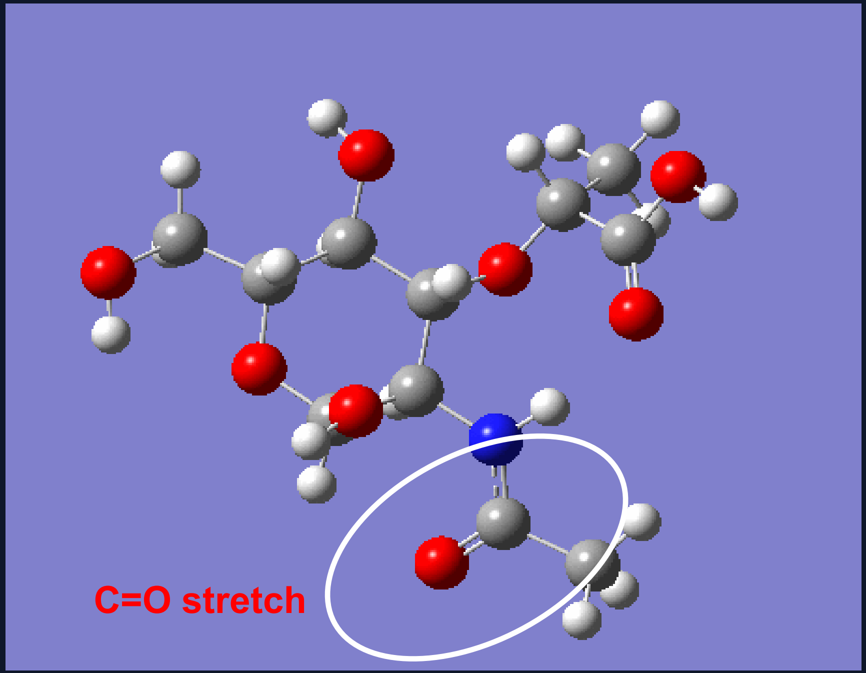
Acquisition Parameter Analysis

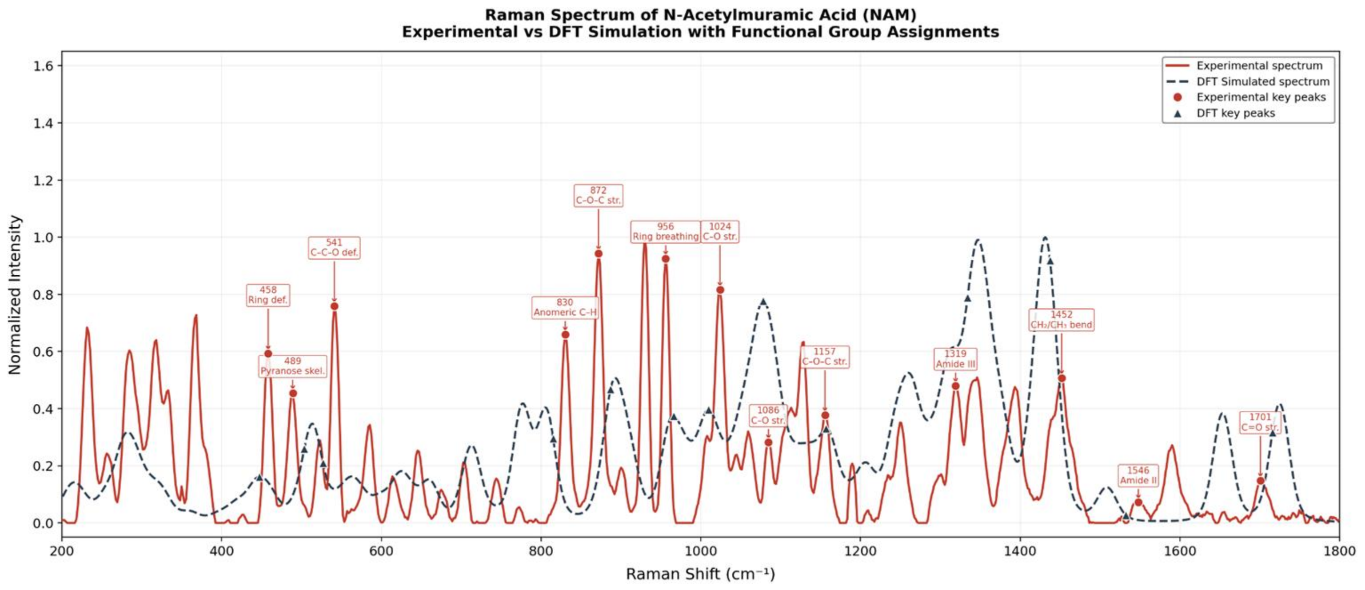


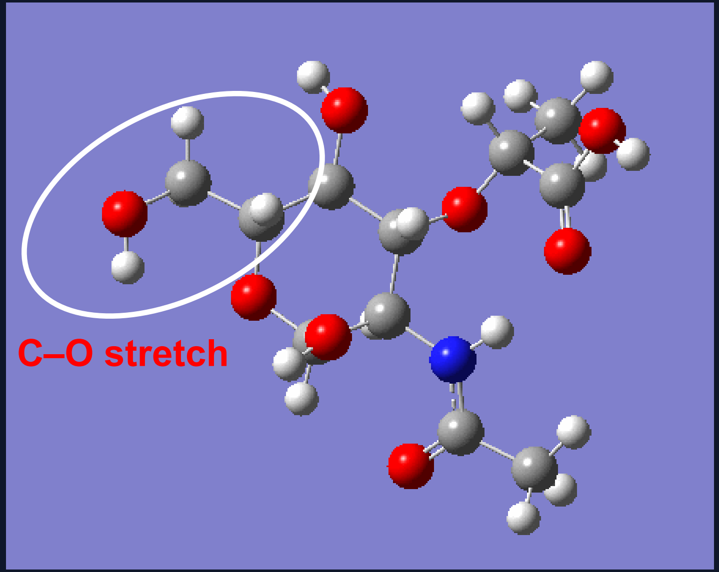


Raman Peak Assignments

Raman Shift	Assignment
1701	C=O stretch
1546	Amide II
1452	CH_2 bending
1086	C–O stretch
872	C–O–C stretch
956	Pyranose ring breathing







Structural Interpretation

Raman spectrum confirms presence of:

Amide Group

1546 cm⁻¹
1319 cm⁻¹

Carbohydrate Backbone

1157–1024 cm⁻¹

Glycosidic Linkage

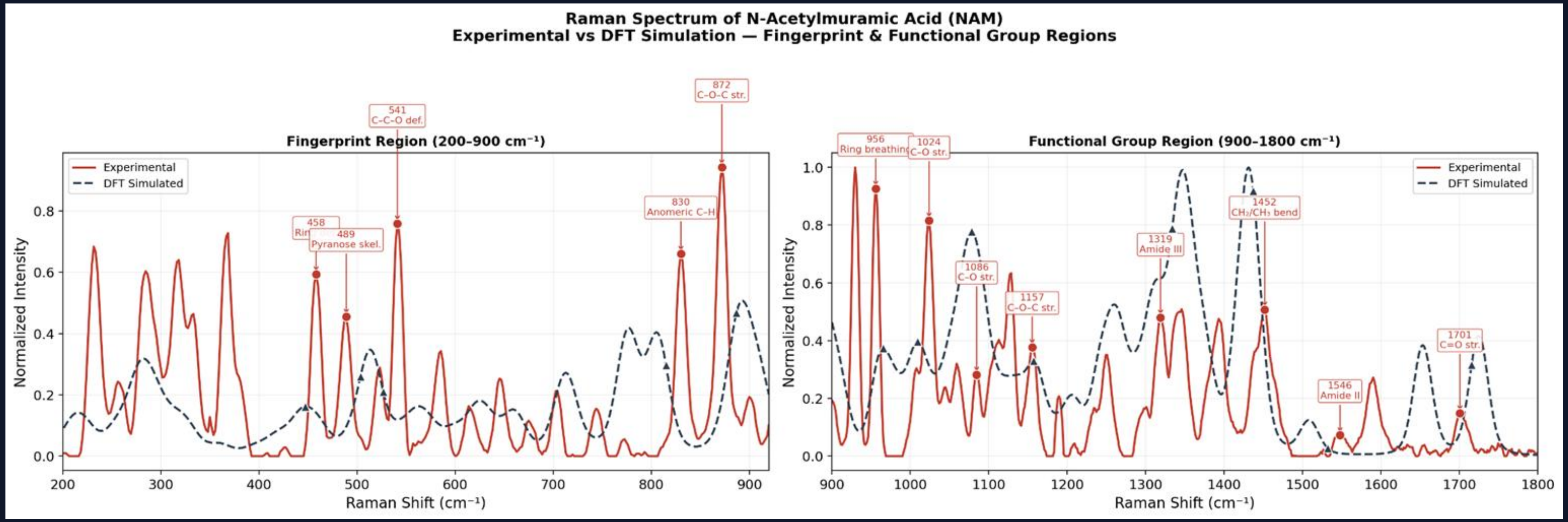
872 cm⁻¹
830 cm⁻¹

Pyranose Ring

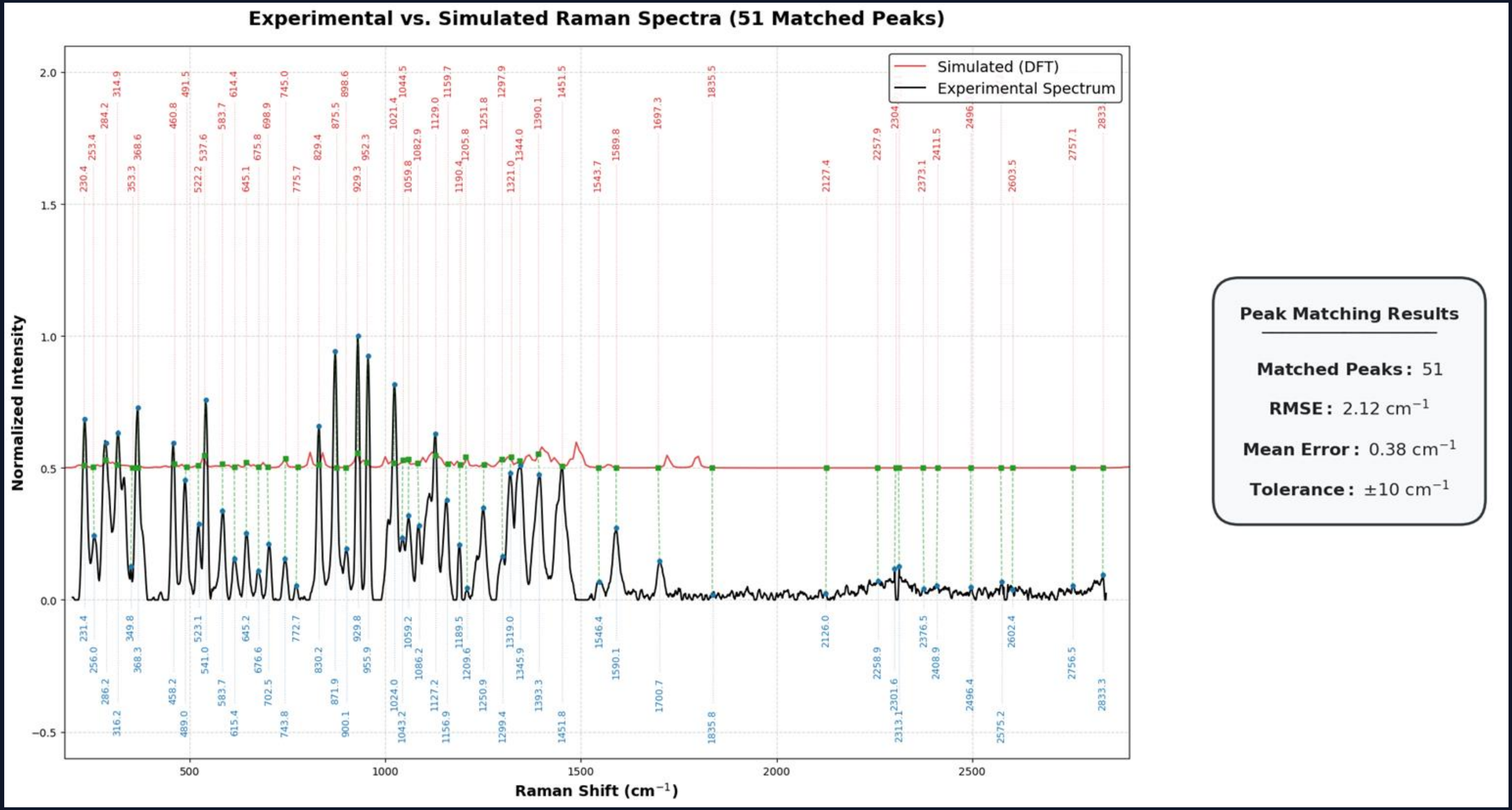
541 cm⁻¹
489 cm⁻¹
458 cm⁻¹

Key findings:

- Strong agreement between **experimental and DFT spectra**
- Raman peaks clearly identify **functional groups**
- DFT accurately predicts **vibrational frequencies**



Experimental vs DFT Raman Spectrum (NAM)



Excellent agreement between experiment and simulation (RMSE ~2 cm^{-1}) validates DFT for biomarker detection

Part III: Lipoteichoic Acid

Challenge in LTA Simulation

- LTA is a **large polymeric biomolecule**
- Full DFT simulation → **computationally impractical**
- Need a **scientifically valid alternative approach**

Component-Based Modeling Strategy

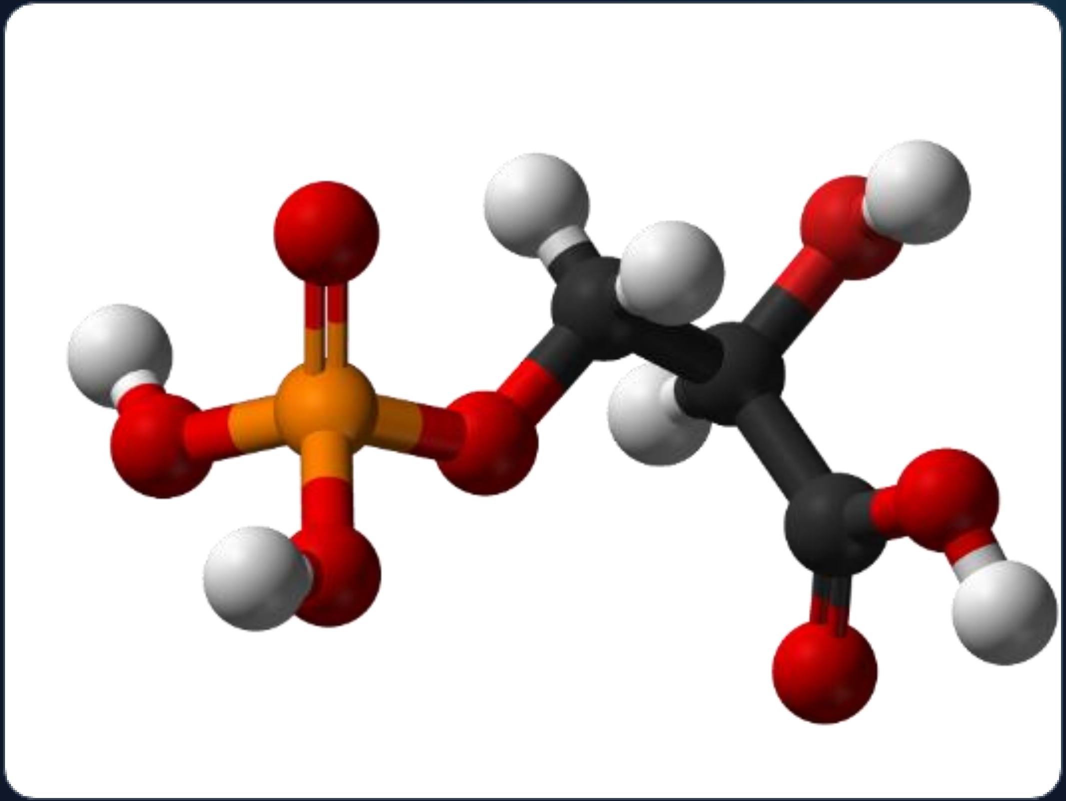
- Decompose LTA into **representative molecular units**
- Simulate each component using DFT
- Combine spectral contributions to reconstruct LTA fingerprint

“Fragment-based modeling is a standard approach for large biomolecules”

Modeling the LTA Fragment

Simplifying the Polymer

- We isolated a representative glycerol-phosphate fragment containing two repeating units.
- This fragment successfully preserves the local chemical environment responsible for key Raman-active backbone vibrations.



Structure of Lipoteichoic Acid

- Lipoteichoic acid consists of three structural components:
  - **Glycolipid membrane anchor**
  - **Poly(glycerol–phosphate) backbone**
  - **Substituent groups** such as D-alanine residues or sugars

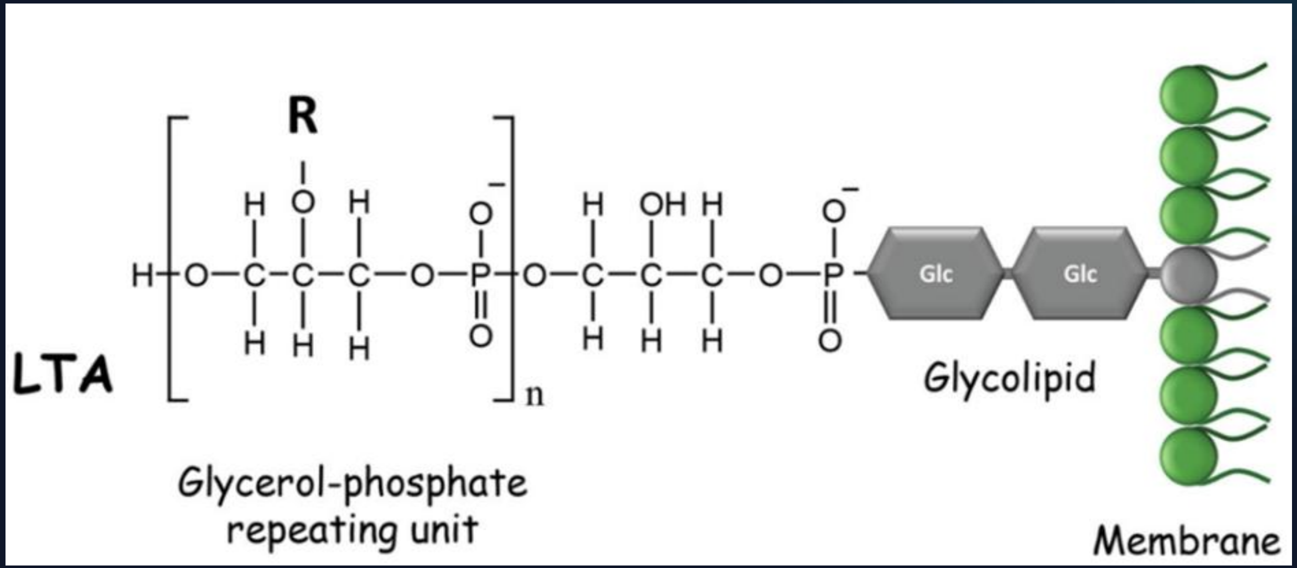
  The backbone is composed of repeating units:

  glycerol – phosphate – glycerol – phosphate

- Typical LTA molecules contain **10–20 repeating glycerol–phosphate units**, producing large polymers with hundreds of atoms.
- Because the full polymer is **computationally expensive for quantum simulations**, smaller molecular fragments are often used to represent the **local chemical environment responsible for Raman-active vibrations**.

Lipoteichoic acid (LTA) is a key structural component of **Gram-positive bacterial cell walls**. It is anchored to the membrane through a glycolipid and extends outward as a polymer of **repeating glycerol–phosphate units**.

General structure:



Lipid anchor – (Gro–P)$_n$

where:

Gro = glycerol
P = phosphate

LTA contributes to:

- bacterial adhesion
- cell wall maintenance
- activation of host immune responses through receptors such as **Toll-like receptor 2**

Raman Spectroscopy and DFT

Raman spectroscopy is a vibrational technique that provides **molecular fingerprints** based on characteristic vibrational modes.

Advantages:

- label-free detection
- minimal sample preparation
- useful for identifying bacterial components

However, experimental Raman spectra often contain **overlapping bands and complex vibrational couplings**.

To improve interpretation, **Density Functional Theory (DFT)** simulations are used to:

- calculate **vibrational frequencies**
- determine **Raman activities**
- assign experimental bands to **specific molecular vibrations**

Objective of this work

- Model a glycerol–phosphate fragment representing the LTA backbone
- Calculate its Raman spectrum using DFT
- Identify characteristic vibrational modes of phosphate and glycerol groups

Molecular Fragment and Structure Construction

To model the LTA backbone, a representative **glycerol–phosphate fragment** containing two repeating units was used:

$HO–CH_2–CH(OH)–CH_2–O–PO_3–O–CH_2–CH(OH)–CH_2–OH$

Key functional groups included:

Functional Group	Role in Raman Spectrum
Phosphate (PO_4)	Strong Raman-active vibrations
C–O–P linkage	Backbone vibrational modes
Hydroxyl groups	Hydrogen bonding interactions
Glycerol backbone	Carbon–oxygen vibrations

The structure was generated using **Avogadro** from the SMILES string:

OP(=O)(O)OCC(O)COP(=O)(O)OCC(O)CO

The molecule was then **geometry optimized using the Universal Force Field (UFF)** to obtain a reasonable starting structure before quantum calculations.

DFT Simulation Method

Quantum-chemical calculations were performed using **Gaussian 16**.

The simulation included:

- geometry optimization
- vibrational frequency calculation
- Raman activity calculation

Parameter	Value
Functional	B3LYP
Basis set	6-31G(d)
Solvent model	SMD (water)

Gaussian keywords used:

```
# B3LYP/6-31G(d) Opt Freq Raman SCF=Tight Int=UltraFine SCRF=(SMD,Solvent=Water)
```

Explanation:

- **SCF=Tight** ensures strict electronic convergence
- **Int=UltraFine** improves numerical accuracy
- **SCRF=SMD** models the aqueous environment in which LTA normally exists

DFT frequencies typically **overestimate vibrational frequencies**, so a scaling factor is applied.

Scaled frequency = frequency × 0.96

Raman Spectrum and Vibrational Assignments

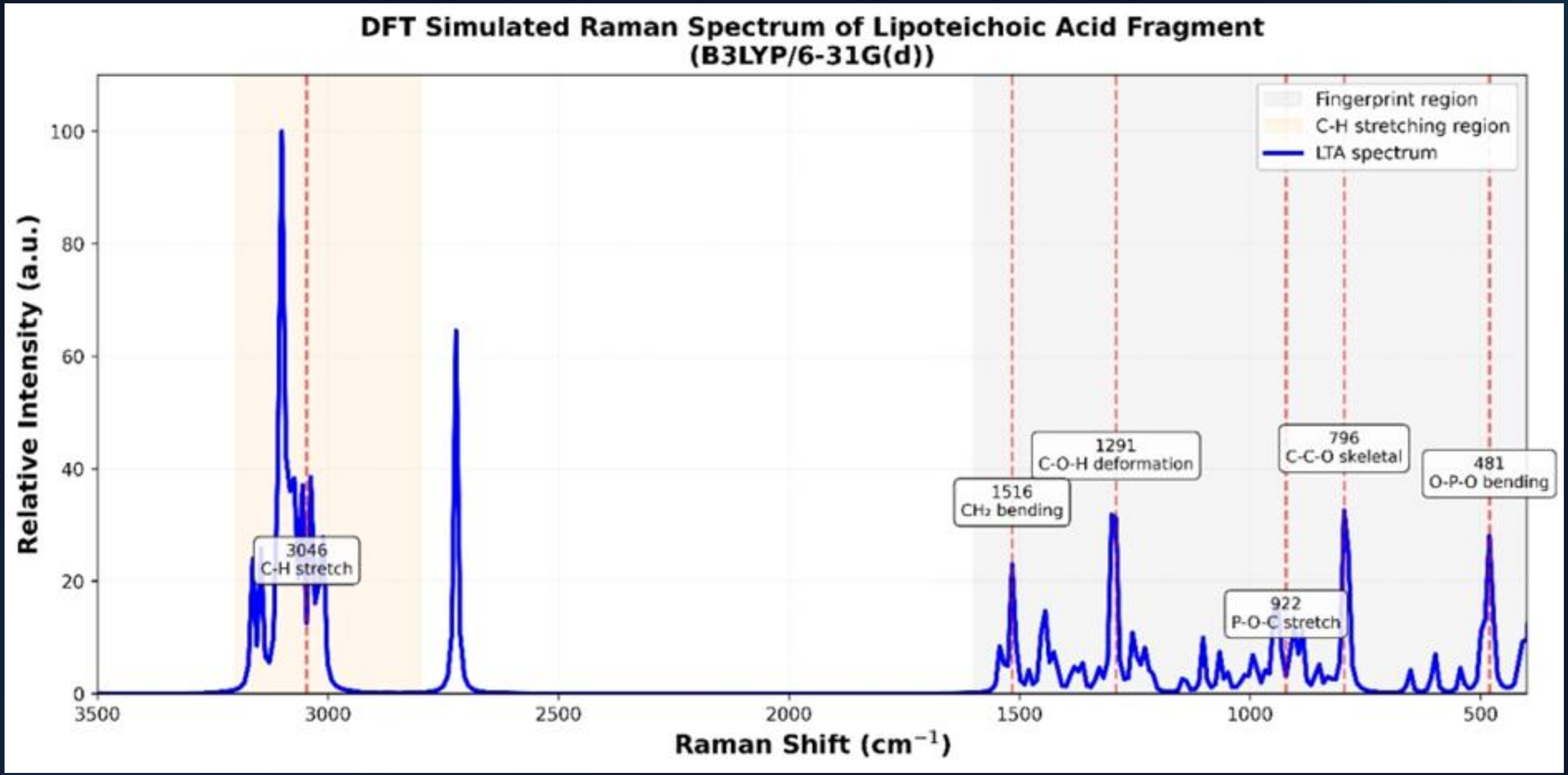
The DFT calculation produced Raman-active vibrational modes corresponding to motions of the phosphate groups and glycerol backbone.

Raman Shift (cm^{-1})	Assignment
481	O–P–O bending
922	P–O–C stretching
1291	O–H deformation
3046–3091	C–H stretching
3712	O–H stretching

Most characteristic information appears in the **fingerprint region (500–1500 cm^{-1})** where phosphate-related vibrations dominate.

Interpretation:

- **481 cm^{-1}** → phosphate bending vibration
- **922 cm^{-1}** → phosphate ester stretching
- **1291 cm^{-1}** → hydroxyl deformation
- **3046–3091 cm^{-1}** → glycerol C–H stretching
- **3712 cm^{-1}** → weakly hydrogen-bonded O–H stretching


DFT Simulated Raman Spectrum of Lipoteichoic Acid Fragment
(B3LYP/6-31G(d))
Relative Intensity (a.u.)
Raman Shift (cm^{-1})
Fingerprint region
C-H stretching region
LTA spectrum
3046
C-H stretch
1516
CH₂ bending
1291
C-O-H deformation
922
P-O-C stretch
796
C-C-O skeletal
481
O-P-O bending
0
20
40
60
80
100
3500
3000
2500
2000
1500
1000
500

Limitations

The model represents only a **small fragment of lipoteichoic acid** and does not include:

- lipid anchors
- substituent groups
- full polymer chain

Therefore, the simulated spectrum mainly reflects **backbone vibrations**.

Conclusion

DFT simulations successfully reproduced the **Raman vibrational modes of a glycerol–phosphate fragment representing the LTA backbone**.

Key findings:

- strong Raman bands originate from **phosphate bending and stretching modes**
- glycerol backbone contributes to **C–H and O–H vibrations**
- DFT calculations provide a useful tool for **interpreting Raman spectra of bacterial cell wall components**

This approach helps improve **spectroscopic identification of bacterial biomarkers**.

Modeling the LTA Substituent: D-Alanine

Why D-Alanine?

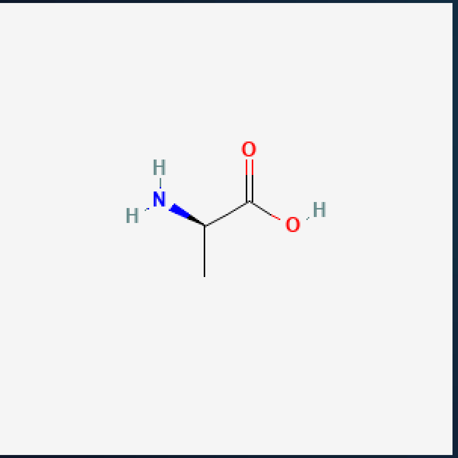
- Represents **amino substituent in LTA**
- Responsible for **surface charge regulation**
- Important for **Gram-positive bacterial identification**

Molecular Characteristics

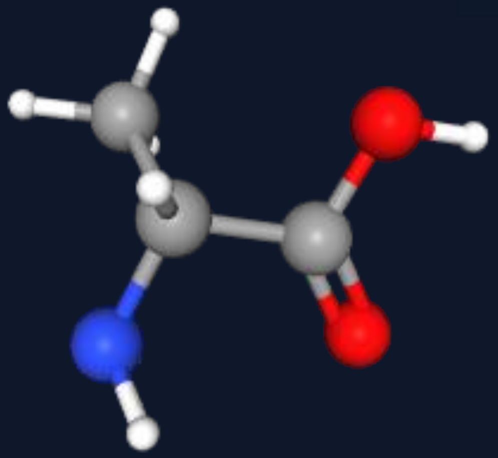
- Exists in **zwitterionic form**: $NH_3^+ – COO^-$
- Contains:
  - Amino group
  - Carboxylate group
  - Methyl group

DFT Simulation Method

- Geometry optimization + Raman calculation
- Functional: B3LYP
- Basis set: 6-31G(d)
- Solvent: SMD (water)



2D Structure



3D Structure

Key Raman Vibrations

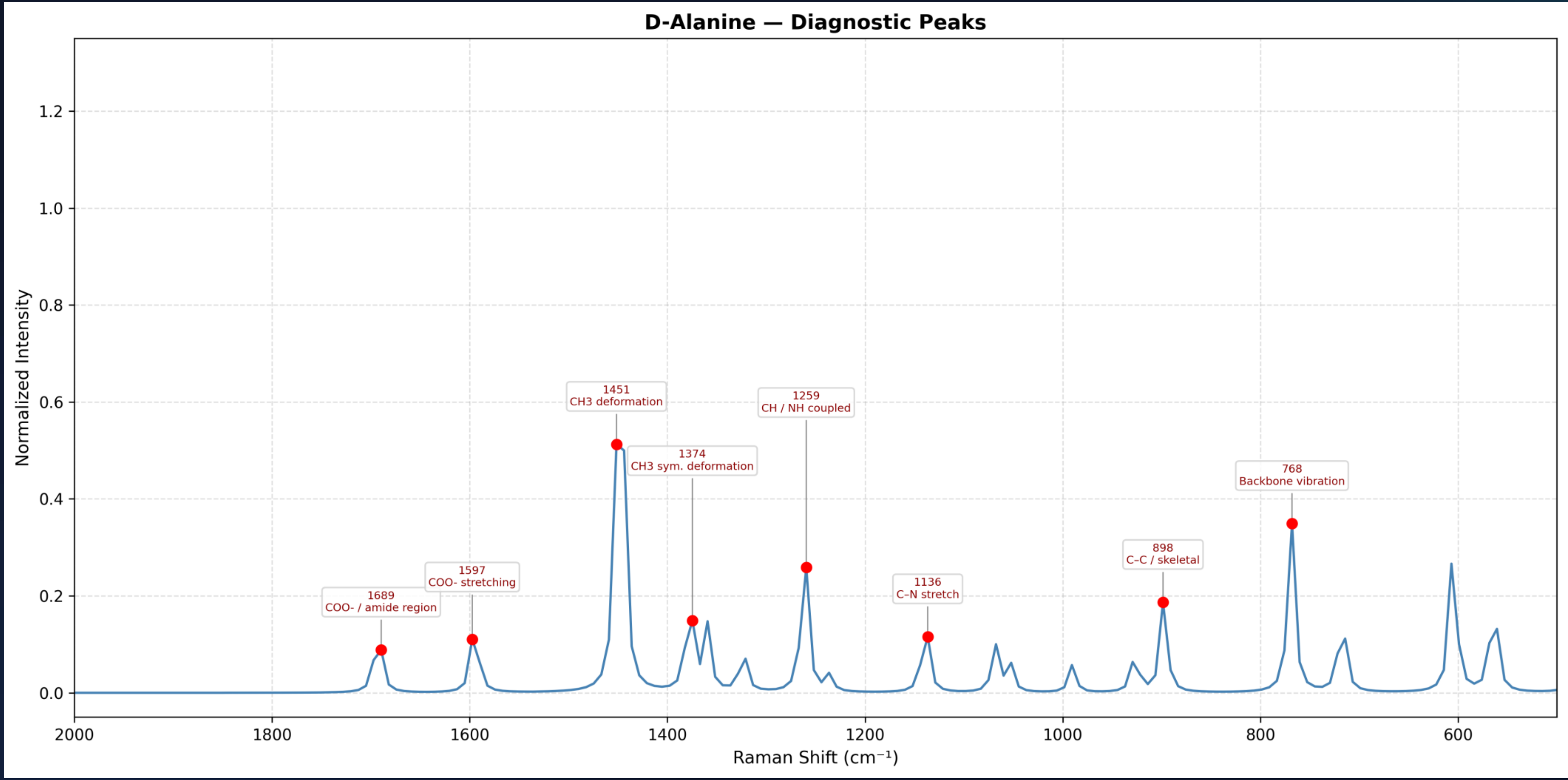
Raman Shift (cm^{-1})	Assignment
~1500	NH_3^+ symmetric deformation
~1560–1600	COO^- stretching
~1440	CH_3 deformation
~1200	C–N stretching

Interpretation

- NH_3^+ and COO^- peaks confirm **zwitterionic amino acid**
- Provides **amino signature in LTA spectrum**

Conclusion

D-Alanine contributes **amino vibrational features**, especially in the **1500–1600 cm^{-1} region**, enabling identification of amino substituents in LTA.



D-Alanine — Diagnostic Peaks
Normalized Intensity
Raman Shift (cm⁻¹)
1.2
1.0
0.8
0.6
0.4
0.2
0.0
2000
1800
1600
1400
1200
1000
800
600
1689
COO- / amide region
1597
COO- stretching
1451
CH3 deformation
1374
CH3 sym. deformation
1259
CH / NH coupled
1136
C–N stretch
898
C–C / skeletal
768
Backbone vibration

Modeling the LTA Substituent: GlcNAc

Why GlcNAc?

- Represents **carbohydrate component**
- Present in **bacterial cell wall structures**
- Provides strong **Raman-active sugar signals**

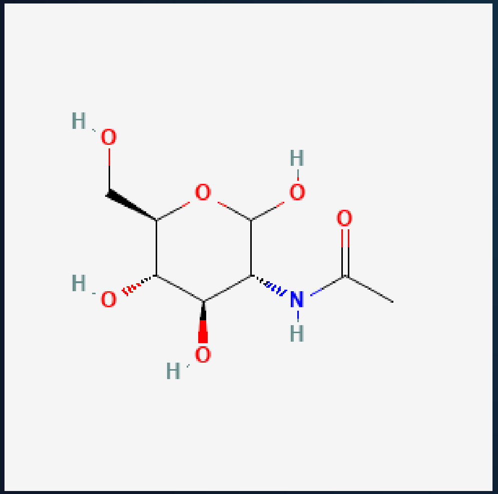
Molecular Characteristics

- Pyranose ring structure
- Contains:

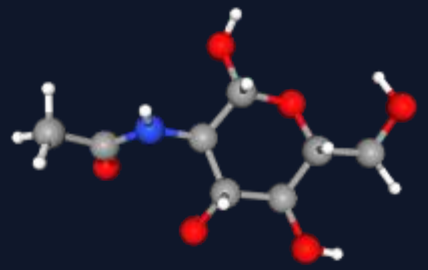
  Multiple hydroxyl groups (–OH)
  Acetamide group (–NHCO–)
  C–O–C linkages

DFT Simulation Method

- Geometry optimization + Raman calculation
- Functional: B3LYP
- Basis set: 6-31G(d)
- Solvent: SMD (water)



2D Structure



3D Structure

Key Raman Vibrations

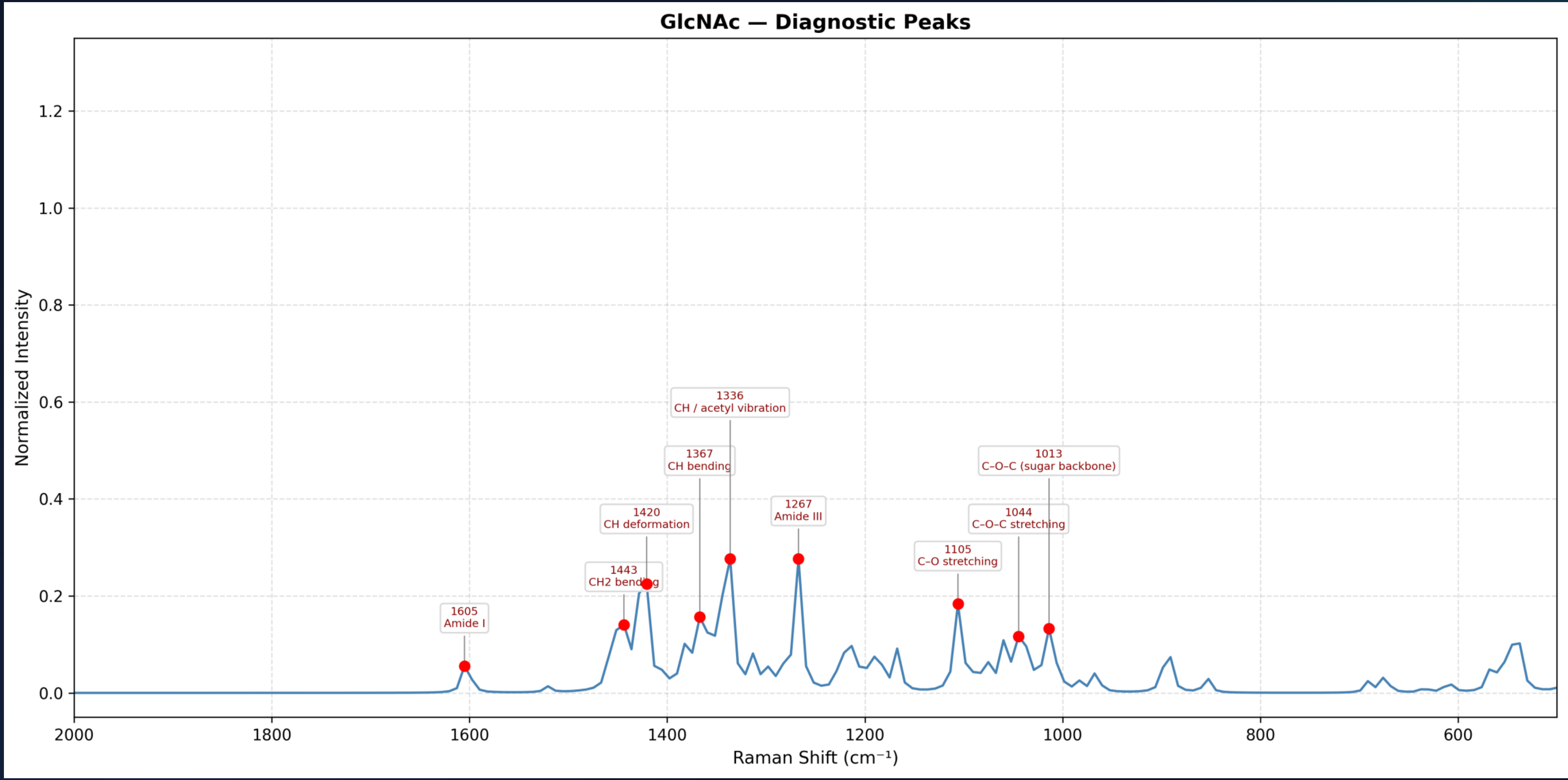
Raman Shift (cm^{-1})	Assignment
~900–1100	C–O–C stretching
~1250–1300	Amide III
~1650	Amide I
~2800–3000	C–H stretching

Interpretation

- Strong **carbohydrate fingerprint region (900–1100 cm^{-1})**
- Amide peaks confirm **acetyl functionality**

Conclusion

GlcNAc contributes **carbohydrate and amide signatures**, forming a major part of the LTA Raman fingerprint.



GlcNAc — Diagnostic Peaks
Normalized Intensity
Raman Shift (cm⁻¹)
0.0
0.2
0.4
0.6
0.8
1.0
1.2
2000
1800
1600
1400
1200
1000
800
600
1605
Amide I
1443
CH2 bending
1420
CH deformation
1367
CH bending
1336
CH / acetyl vibration
1267
Amide III
1105
C–O stretching
1044
C–O–C stretching
1013
C–O–C (sugar backbone)

Raman Spectroscopic Analysis of Lipoteichoic Acid (LTA)

Content:

Sample: Lipoteichoic Acid (LTA) from *Staphylococcus aureus*
Technique: Raman Spectroscopy (785 nm laser)

Objective:

To identify functional groups of LTA
To compare experimental spectra with expected molecular regions

Approach:

Power variation study (40%, 50%, 60%, 70%)
Baseline correction + smoothing

Experimental Methodology & Data Processing

Content:

- Raman spectra collected in **500–2000 cm^{-1} region**
- Instrument parameters:

  Integration time: 10 sec
  Accumulations: 10
  Laser power: 40%, 50%, 60%, 70%

Data Processing Steps:

- Dark subtraction
- Baseline correction (ALS method)
- Smoothing (Savitzky–Golay filter)

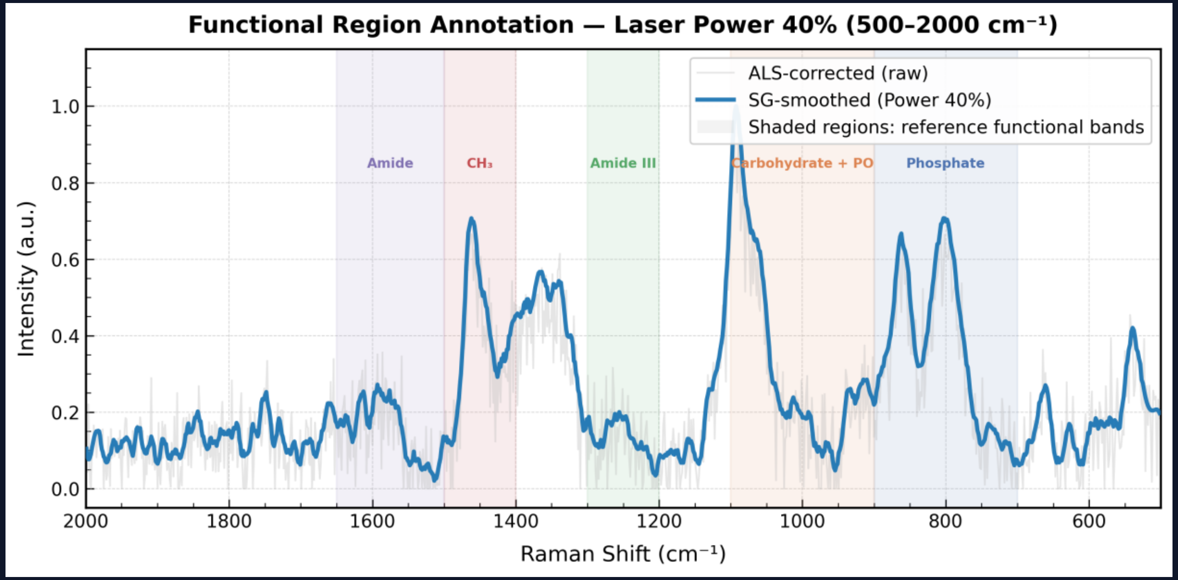
Visualization:

•Functional regions highlighted using **color-coded bands**
•Region-based interpretation instead of exact peak assignment

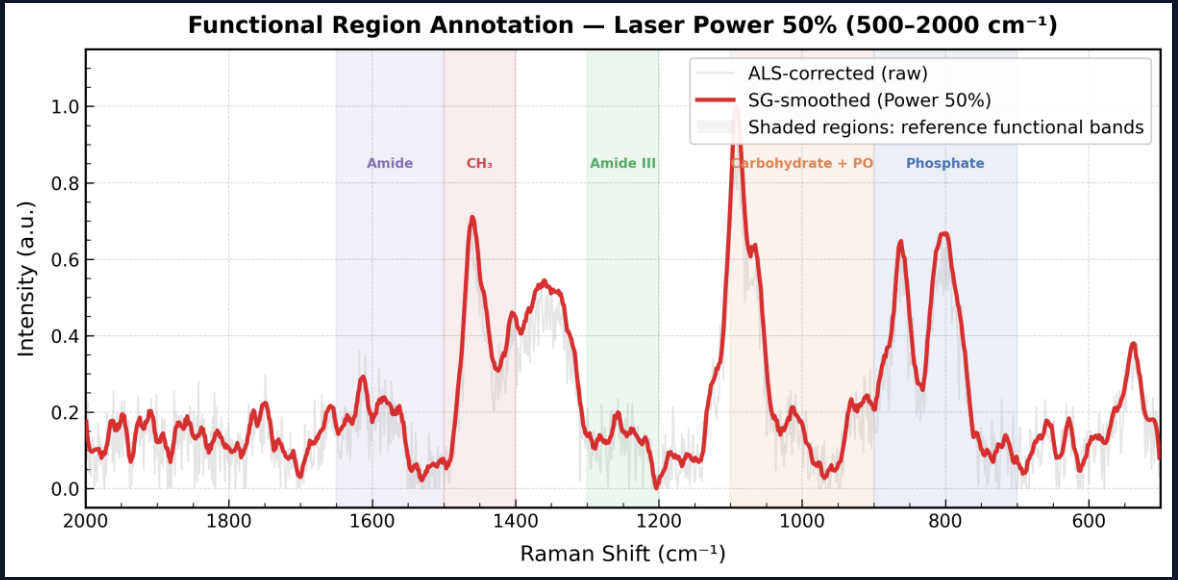
Functional Region Identification (LTA Components)

Region (cm^{-1})	Assignment	Component
700–900	P–O–C vibration	Phosphate backbone
900–1100	C–O–C + PO stretch	Carbohydrate (GlcNAc)
1200–1300	Amide III	GlcNAc acetyl group
1400–1500	CH_3 deformation	D-Alanine
1500–1650	Amide / NH_3^+ / COO^-	Amino + peptide region

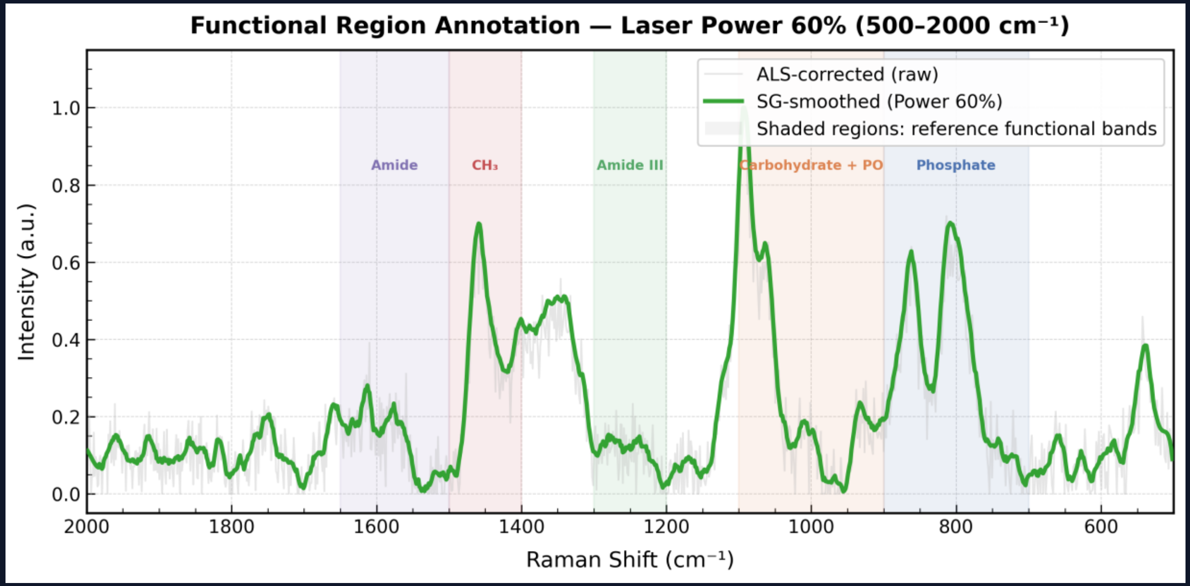
These regions correspond to **key structural components of LTA**



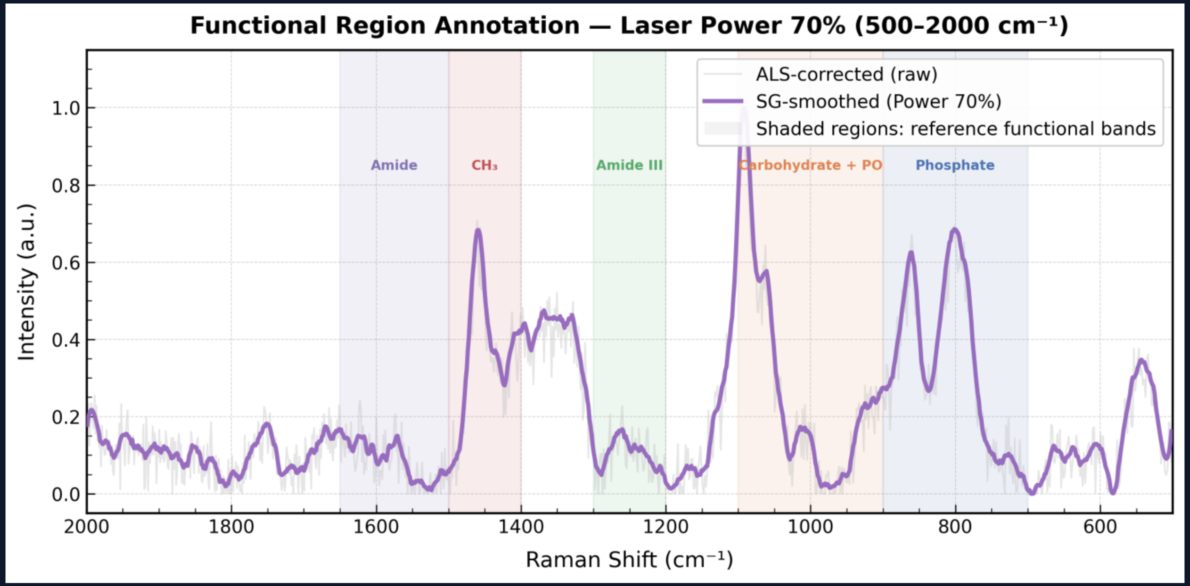
Functional Region Annotation — Laser Power 40% (500–2000 cm⁻¹)
ALS-corrected (raw)
SG-smoothed (Power 40%)
Shaded regions: reference functional bands
Amide
CH₃
Amide III
Carbohydrate + PO
Phosphate
Intensity (a.u.)
0.0
0.2
0.4
0.6
0.8
1.0
2000
1800
1600
1400
1200
1000
800
600
Raman Shift (cm⁻¹)



Functional Region Annotation — Laser Power 50% (500–2000 cm⁻¹)
ALS-corrected (raw)
SG-smoothed (Power 50%)
Shaded regions: reference functional bands
Amide
CH₃
Amide III
Carbohydrate + PO
Phosphate
Intensity (a.u.)
0.0
0.2
0.4
0.6
0.8
1.0
2000
1800
1600
1400
1200
1000
800
600
Raman Shift (cm⁻¹)

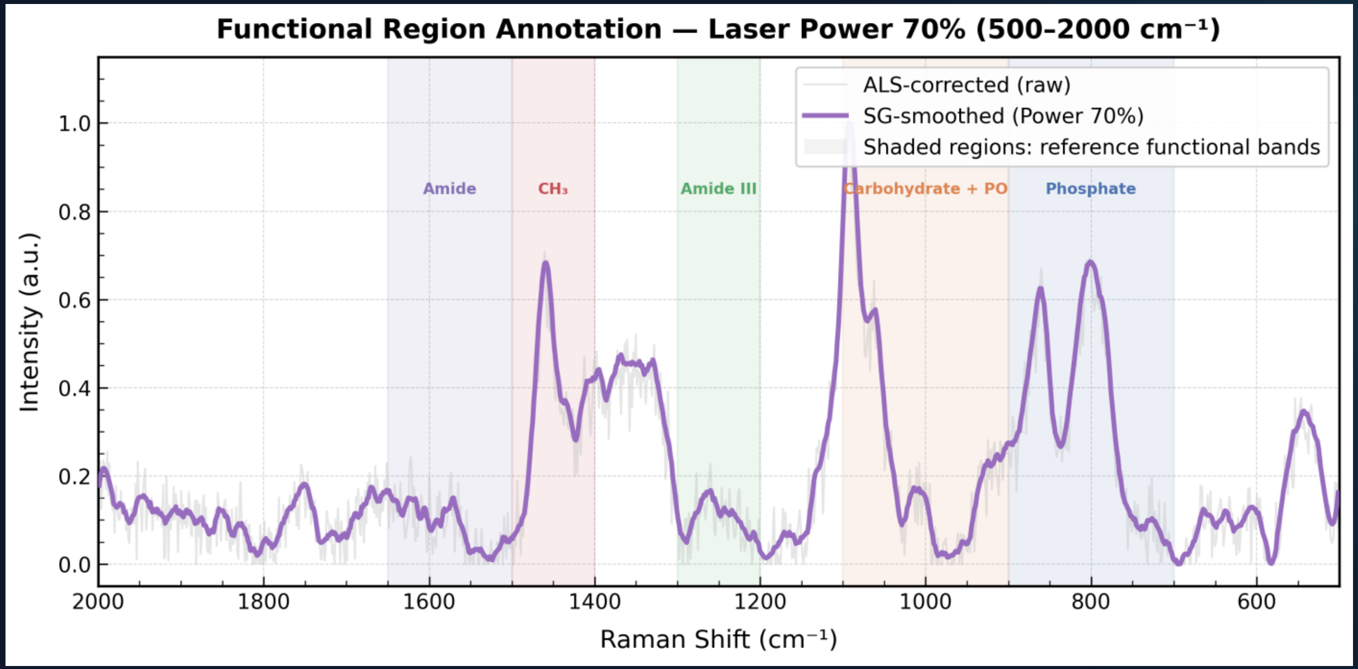


Functional Region Annotation — Laser Power 60% (500–2000 cm⁻¹)
ALS-corrected (raw)
SG-smoothed (Power 60%)
Shaded regions: reference functional bands
Amide
CH₃
Amide III
Carbohydrate + PO
Phosphate
Intensity (a.u.)
0.0
0.2
0.4
0.6
0.8
1.0
2000
1800
1600
1400
1200
1000
800
600
Raman Shift (cm⁻¹)

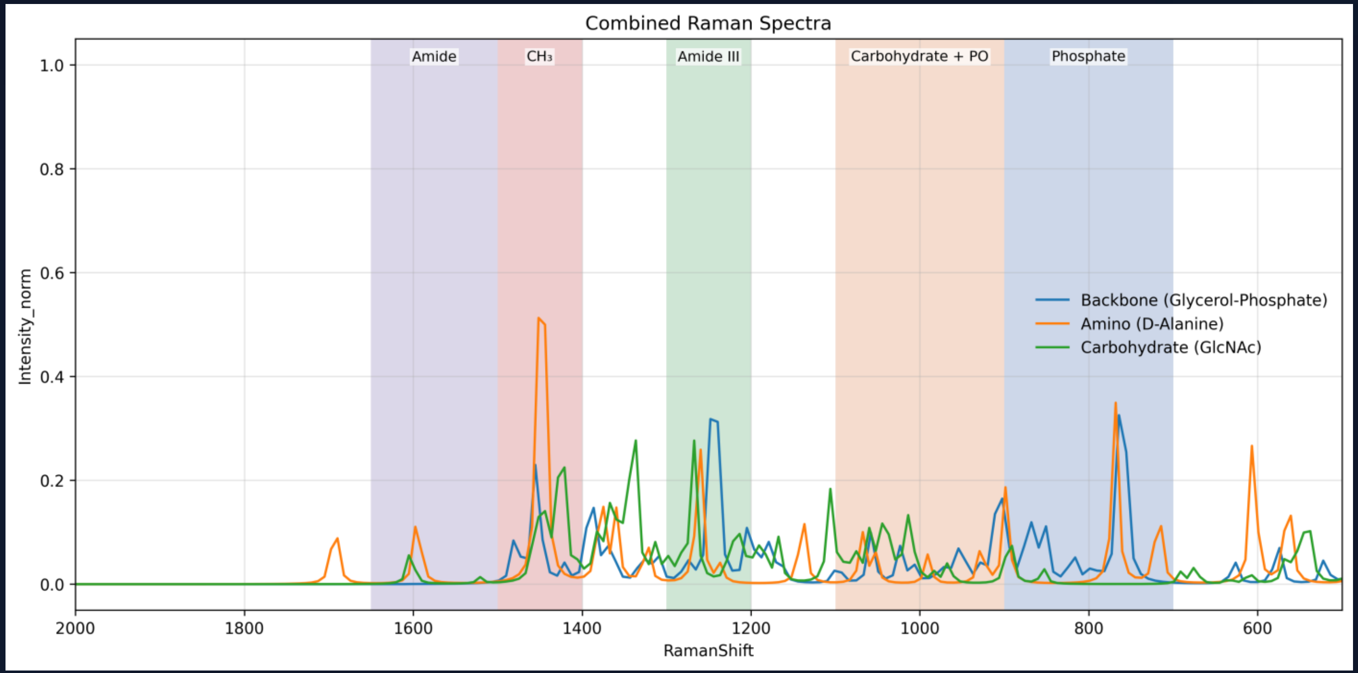


Functional Region Annotation — Laser Power 70% (500–2000 cm⁻¹)
ALS-corrected (raw)
SG-smoothed (Power 70%)
Shaded regions: reference functional bands
Amide
CH₃
Amide III
Carbohydrate + PO
Phosphate
Intensity (a.u.)
0.0
0.2
0.4
0.6
0.8
1.0
2000
1800
1600
1400
1200
1000
800
600
Raman Shift (cm⁻¹)

Experimental Data

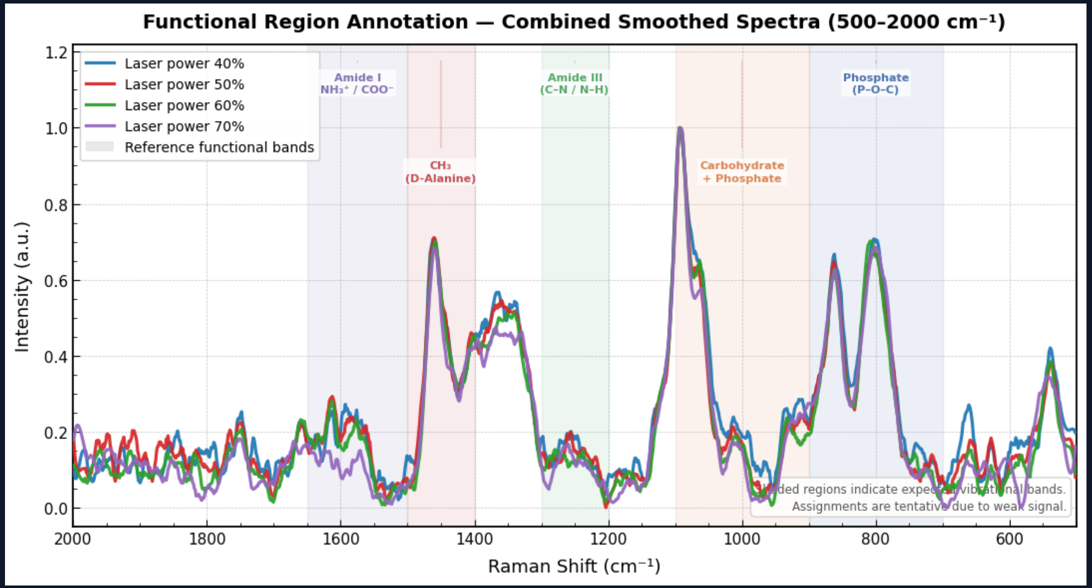


Simulation Data



Observations from Raman Spectra

- All spectra show **broad and overlapping peaks**
- Key regions are **present but not sharply resolved**
- Strong signal observed in:
  - ~1100 cm^{-1} → phosphate/carbohydrate region
  - ~800–900 cm^{-1} → phosphate backbone
- Moderate signal in:
  - ~1400–1500 cm^{-1} → CH_3 (D-Alanine)
- Weak/merged signal in:
  - Amide regions (1200–1300, 1500–1650 cm^{-1})
- Indicates **presence of LTA functional groups but low peak resolution**



Conclusion & Interpretation

Content:

Functional regions of LTA were successfully identified

Experimental spectra show:

1. Broad peaks due to complex biomolecule structure
2. Overlapping contributions from multiple components

Region-based analysis confirms:

- Phosphate backbone
- Carbohydrate structure
- Amino acid (D-Alanine) presence

Limitation:

Peaks are not sharp due to:

- Weak Raman scattering
- Sample form (powder)

Future Work & Next Steps

1. Experimental Extension (NAM)
- Serial dilution of NAM samples
- Raman data collection at different concentrations
- Study of signal variation with concentration

2. Quantitative Analysis
- Comparative analysis of peak intensity vs concentration
- Evaluation of detection limits
- Signal-to-noise variation across dilution levels

3. LTA Work
- Serial dilution of LTA samples
- Raman data collection at different concentrations
- Study of signal variation with concentration

4. LPS Work
- DFT simulation of LPS components (Lipid A / phosphate groups)
- Identification of characteristic Raman peaks
- Literature comparison for validation

Key Questions from Last Meeting

1. Detection vs Quantification
Question: Can Raman/SERS quantify bacteria or only detect presence?
Answer: Raman/SERS primarily provides qualitative detection through molecular fingerprints. However, with calibration methods such as serial dilution and intensity–concentration mapping, it can be extended to semi-quantitative estimation of bacterial load.

2. Speed vs Culture
Question: How is Raman better than culture if culture provides accurate results?
Answer: Culture methods typically take 48–72 hours, whereas Raman/SERS can provide results within minutes. Even if quantification is approximate, rapid detection enables early clinical decision-making, which is critical in acute conditions.

3. Infection Stage Estimation
Question: Can you determine whether infection is low or severe?
Answer: Yes. By correlating Raman intensity with bacterial concentration, infection levels can be classified as low, medium, or high, providing an indication of disease severity.

4. Accuracy of Quantification
Question: How reliable is Raman-based quantification?
Answer: Raman quantification can be challenging due to signal variability. However, with proper calibration and controlled experimental conditions, it can provide clinically useful approximate estimates.

5. Dead vs Alive Bacteria
Question: Raman detects dead bacteria as well—how do you address this limitation?
Answer: Raman detects molecular signatures regardless of viability. To address this, time-dependent measurements can be used. If signal intensity increases over time, it indicates active bacterial growth rather than residual presence.

6. Growth Monitoring
Question: Can you track whether bacteria are multiplying?
Answer: Yes. By acquiring Raman spectra at multiple time points (e.g., 0, 30, and 60 minutes), trends can be observed. Increasing intensity indicates active growth, while stable or decreasing intensity suggests inactive or controlled infection.

7. Predictive Modeling
Question: Can we build a model to estimate bacterial quantity?
Answer: Yes. Models can be developed to map spectral features to concentration. Additionally, machine learning approaches such as regression or neural networks can improve prediction accuracy.

8. Clinical Usefulness
Question: How will this approach help clinicians?
Answer: It provides rapid information on both the presence and approximate bacterial load, enabling clinicians to decide whether treatment is necessary and how aggressive it should be.

9. Importance of Quantification
Question: Why is quantification important clinically?
Answer: Bacterial load correlates with disease severity. Low levels may not require antibiotics, whereas high levels indicate active infection requiring intervention. Therefore, quantification supports clinical decision-making.

10. Clinical Validation
Question: What level of estimation is useful for clinicians?
Answer: This requires validation with clinicians to determine what level of accuracy is clinically actionable and what thresholds should guide treatment decisions.

Questions & Discussion

Thank you